A quantitative Kohler–Jobin argument through level sets

1 The general idea

For a bounded open set $D \subset \mathbb{R}^n$, let

$$\lambda_1(D) = \min_{0 \neq \varphi \in H_0^1(D)} \frac{\displaystyle\int_D |\nabla \varphi|^2 \, \mathrm{d}x}{\displaystyle\int_D \varphi^2 \, \mathrm{d}x}$$

be the first Dirichlet eigenvalue, and let

$$T(D) = \max_{0 \neq \varphi \in H_0^1(D)} \frac{\displaystyle\left(\int_D \varphi \, \mathrm{d}x\right)^2}{\displaystyle\int_D |\nabla \varphi|^2 \, \mathrm{d}x}$$

be the torsional rigidity.

If w_D denotes the torsion function,

$$-\Delta w_D = 1 \quad \text{in } D, \qquad w_D \in H_0^1(D),$$

then

$$T(D) = \int_D w_D \, \mathrm{d}x = \int_D |\nabla w_D|^2 \, \mathrm{d}x.$$

The Kohler–Jobin inequality states that

$$\lambda_1(\Omega) T(\Omega)^{\frac{2}{n+2}}$$

is minimized by balls.

The aim of this note is to explain how a small Kohler–Jobin deficit can be converted into a small Faber–Krahn deficit on a suitable superlevel set of the first eigenfunction.

The architecture is closely related to the one appearing in a quantitative version of Chiti's inequality: one first controls the level structure of the eigenfunction, then chooses a level whose superlevel set has the correct volume, and finally uses the truncation of the eigenfunction as a competitor for the first eigenvalue of this superlevel set.

The main difference lies in the mechanism producing the initial control on the levels. In the Chiti setting this comes from the single-crossing property of rearrangements. Here it comes from the Kohler–Jobin rearrangement and the modified torsional rigidity.

2 Normalization

For balls we write

$$T(B_R) = \tau_n R^{n+2}, \qquad \lambda_1(B_R) = \Lambda_n R^{-2},$$

where

$$\tau_n := T(B_1), \qquad \Lambda_n := \lambda_1(B_1).$$

Since

$$T(rD) = r^{n+2}T(D), \qquad \lambda_1(rD) = r^{-2}\lambda_1(D),$$

the quantity

$$\lambda_1(D)T(D)^{\frac{2}{n+2}}$$

is scale invariant.

We may therefore normalize

$$T(\Omega) = T(B_1) = \tau_n.$$

We write

$$\lambda_1(\Omega) = \Lambda_n(1+\delta),$$

so that

$$\delta = \frac{\lambda_1(\Omega)}{\lambda_1(B_1)} - 1$$

is the normalized Kohler–Jobin deficit.

Let $u > 0$ be the first eigenfunction of Ω, normalized by

$$\|u\|_{L^2(\Omega)} = 1.$$

Thus

$$-\Delta u = \lambda_1(\Omega)u \quad \text{in } \Omega.$$

We shall show that a small value of δ produces a suitable superlevel set of u with small Faber–Krahn deficit.

3 Modified torsional rigidity

The Kohler–Jobin rearrangement does not use the full torsional problem directly. Instead, one restricts the admissible functions to functions depending only on u, namely functions of the form

$$g(u).$$

This gives the modified torsional rigidity

$$T_{\mathrm{mod}}(\Omega; u).$$

Since the admissible class is smaller than $H_0^1(\Omega)$,

$$T_{\mathrm{mod}}(\Omega; u) \le T(\Omega) = \tau_n.$$

Let B_r be the ball such that

$$T(B_r) = T_{\mathrm{mod}}(\Omega; u). \tag{1}$$

Since $T(B_r) = \tau_n r^{n+2}$, we have

$$r \le 1.$$

The importance of this ball is that it is precisely the comparison ball naturally produced by the Kohler–Jobin rearrangement.

4 The Kohler–Jobin rearrangement

Starting from u, the Kohler–Jobin construction produces a radial decreasing function

$$v \in H_0^1(B_r)$$

such that

$$\int_{B_r} |\nabla v|^2 \, \mathrm{d}x = \int_{\Omega} |\nabla u|^2 \, \mathrm{d}x = \lambda_1(\Omega), \tag{2}$$

and, in particular,

$$\int_{B_r} v^2 \, \mathrm{d}x \geq \int_{\Omega} u^2 \, \mathrm{d}x = 1. \tag{3}$$

Set

$$A := \int_{B_r} v^2 \, \mathrm{d}x.$$

Then

$$A \geq 1.$$

Using v as a competitor in the Rayleigh quotient of B_r, we obtain

$$\lambda_1(B_r) \leq \frac{\displaystyle\int_{B_r} |\nabla v|^2 \, \mathrm{d}x}{\displaystyle\int_{B_r} v^2 \, \mathrm{d}x} = \frac{\lambda_1(\Omega)}{A}.$$

Since

$$\lambda_1(B_r) = \Lambda_n r^{-2}, \qquad \lambda_1(\Omega) = \Lambda_n(1+\delta),$$

we get

$$\frac{A}{r^2} \leq 1 + \delta.$$

Since $A \geq 1$ and $r \leq 1$, it follows that

$$0 \leq A - 1 \leq \delta \tag{4}$$

and

$$(1+\delta)^{-1/2} \leq r \leq 1. \tag{5}$$

Thus the first quantitative information supplied by the Kohler–Jobin deficit is the estimate

$$A - 1 \leq \delta.$$

The quantity $A - 1$ measures the L^2-mass gained by the Kohler–Jobin rearrangement. Moreover, from (5),

$$0 \leq T(\Omega) - T_{\mathrm{mod}}(\Omega; u) = \tau_n(1 - r^{n+2}) \leq C_n \delta.$$

5 The superlevel structure

Set

$$M := \|u\|_{L^\infty(\Omega)}$$

and define

$$\mu(t) := |\{u > t\}|, \qquad 0 \le t \le M.$$

We also write

$$\Omega_t := \{u > t\}.$$

For each t, consider the truncation

$$(u - t)_+.$$

We define

$$\mathcal{T}(t) := T_{\mathrm{mod}}\big(\Omega_t; (u-t)_+\big).$$

The point of this definition is that $\mathcal{T}(t)$ measures the modified torsional rigidity remaining above the level t.

Let

$$F(t) := \int_{\{u>t\}} u \, \mathrm{d}x.$$

Integrating

$$-\Delta u = \lambda_1(\Omega) u$$

over the superlevel set $\{u > t\}$ gives, for almost every t,

$$\int_{\{u=t\}} |\nabla u| \, \mathrm{d}\mathcal{H}^{n-1} = \lambda_1(\Omega) F(t). \tag{6}$$

Using the level-set representation of the modified torsion, one finds

$$\mathcal{T}(t) = \int_t^M \frac{\mu(s)^2}{\lambda_1(\Omega) F(s)} \, \mathrm{d}s. \tag{7}$$

Therefore

$$-\mathcal{T}'(t) = \frac{\mu(t)^2}{\lambda_1(\Omega) F(t)} \tag{8}$$

for almost every t.

6 The comparison volume $\beta(t)$

For every t, define $\beta(t)$ to be the volume of the ball having torsional rigidity $\mathcal{T}(t)$.

Since

$$T(B_\rho) = \tau_n \rho^{n+2}, \qquad |B_\rho| = \omega_n \rho^n,$$

we have

$$\beta(t) = \omega_n \left(\frac{\mathcal{T}(t)}{\tau_n} \right)^{\frac{n}{n+2}}. \tag{9}$$

At $t = 0$,

$$\mathcal{T}(0) = T_{\mathrm{mod}}(\Omega; u) = T(B_r),$$

hence

$$\beta(0) = |B_r|.$$

We set

$$b := |B_r|.$$

A fundamental comparison gives

$$\beta(t) \le \mu(t). \tag{10}$$

This has a simple interpretation. The set Ω_t has volume $\mu(t)$, while $\mathcal{T}(t)$ is no larger than its usual torsional rigidity. By the Saint–Venant inequality, the ball of volume $\mu(t)$ maximizes torsion. Hence the ball having torsion $\mathcal{T}(t)$ cannot have volume larger than $\mu(t)$.

7 The displacement of the levels

We now compare the levels of u with those of the radial function v produced by the Kohler–Jobin rearrangement.

Let

$$\Phi(t)$$

be the level of v corresponding to the level t of u.

The construction gives

$$|\{v > \Phi(t)\}| = \beta(t).$$

Moreover,

$$\Phi'(t) = \frac{\mu(t)}{\beta(t)} \tag{11}$$

for almost every t.

Because of (10),

$$\Phi'(t) \ge 1.$$

We define the displacement function

$$D(t) := \Phi(t) - t.$$

Since

$$\Phi(0) = 0,$$

we have

$$D(0) = 0.$$

Furthermore,

$$D'(t) = \frac{\mu(t)}{\beta(t)} - 1 \ge 0. \tag{12}$$

Thus D is nondecreasing.

This is one of the key structural properties of the construction. Once a positive displacement between the levels of u and those of v has accumulated, it persists at all higher levels.

8 An exact identity for the L^2-gain

By the layer-cake formula,

$$\int_\Omega u^2 \, \mathrm{d}x = 2 \int_0^M t\mu(t) \, \mathrm{d}t.$$

For v,

$$\int_{B_r} v^2 \, \mathrm{d}x = 2 \int_0^\infty s|\{v > s\}| \, \mathrm{d}s.$$

Using the change of variables

$$s = \Phi(t),$$

together with

$$|\{v > \Phi(t)\}| = \beta(t)$$

and

$$\Phi'(t) = \frac{\mu(t)}{\beta(t)},$$

we obtain

$$\begin{aligned} \int_{B_r} v^2 \, \mathrm{d}x &= 2 \int_0^M \Phi(t)\beta(t)\Phi'(t) \, \mathrm{d}t \\ &= 2 \int_0^M \Phi(t)\mu(t) \, \mathrm{d}t. \end{aligned}$$

Subtracting the corresponding formula for u, and recalling $\|u\|_2 = 1$, we find

$$\boxed{A - 1 = 2 \int_0^M D(t)\mu(t) \, \mathrm{d}t.} \tag{13}$$

Combining (13) with (4) gives

$$2 \int_0^M D(t)\mu(t) \, \mathrm{d}t \le \delta. \tag{14}$$

This is the quantitative level-set information hidden in the Kohler–Jobin deficit.

9 The distinguished level $\bar{t}$

We now choose the level at which the superlevel set has exactly the same volume as B_r.

Define

$$\bar{t} := u^*(b), \qquad b = |B_r|. \tag{15}$$

Thus

$$|\{u > \bar{t}\}| = b = |B_r|. \tag{16}$$

Set

$$U := \{u > \bar{t}\}.$$

The reason for choosing $\bar{t}$ in this way is that U and B_r have the same volume, so that the Faber–Krahn inequality, and later its quantitative form, can be applied directly to the pair (U, B_r).

The remaining task is to show that $\bar{t}$ is small when δ is small.

10 Uniform bounds on the eigenfunction

We first collect two elementary estimates.

Using u as a competitor in the variational characterization of torsion,

$$T(\Omega) \geq \frac{\left(\int_\Omega u \, \mathrm{d}x\right)^2}{\int_\Omega |\nabla u|^2 \, \mathrm{d}x}.$$

Since

$$T(\Omega) = \tau_n, \qquad \int_\Omega |\nabla u|^2 \, \mathrm{d}x = \lambda_1(\Omega),$$

we obtain

$$\|u\|^2_{L^1(\Omega)} \leq \tau_n \lambda_1(\Omega). \tag{17}$$

For $\delta \leq 1$, this gives

$$\|u\|_{L^1(\Omega)} \leq C_n.$$

We shall also use the standard L^∞ bound for a L^2-normalized first eigenfunction,

$$\|u\|_{L^\infty(\Omega)} \leq C_n \lambda_1(\Omega)^{n/4} \leq C_n \tag{18}$$

for bounded δ.

11 Quadratic growth of the level displacement

For

$$0 \leq t \leq \bar{t},$$

the definition of $\bar{t}$ gives

$$\mu(t) \geq b. \tag{19}$$

From (5), for $\delta \leq 1$,

$$b = |B_r| \geq c_n > 0.$$

Using (8), (19), and

$$F(t) \leq \|u\|_{L^1(\Omega)},$$

we obtain

$$-\mathcal{T}'(t) = \frac{\mu(t)^2}{\lambda_1(\Omega) F(t)} \geq c_n$$

for

$$0 \leq t \leq \bar{t}.$$

Integrating,

$$\mathcal{T}(0) - \mathcal{T}(t) \geq c_n t. \tag{20}$$

Recall that

$$\beta(t) = \omega_n \left(\frac{\mathcal{T}(t)}{\tau_n}\right)^{\frac{n}{n+2}}.$$

Since $\mathcal{T}(0)$ is uniformly bounded away from zero when δ is small, (20) implies

$$b - \beta(t) \geq c_n t. \tag{21}$$

Now

$$D'(t) = \frac{\mu(t) - \beta(t)}{\beta(t)}.$$

Using $\mu(t) \geq b$,

$$D'(t) \geq \frac{b - \beta(t)}{b} \geq c_n t.$$

Since $D(0) = 0$,

$$\boxed{D(t) \geq c_n t^2, \qquad 0 \leq t \leq \bar{t}.} \tag{22}$$

12 A lower bound on the mass above small levels

For $t \geq 0$, set

$$g_t := (u - t)_+.$$

Pointwise,

$$0 \leq u^2 - g_t^2 \leq 2tu.$$

Hence

$$\|g_t\|_2^2 \geq 1 - 2t\|u\|_1.$$

Using the uniform L^1 bound, there exists

$$t_0 = t_0(n) > 0$$

such that

$$t \leq t_0 \quad \Longrightarrow \quad \|g_t\|_2^2 \geq \frac{1}{2}. \tag{23}$$

Moreover,

$$\|g_t\|_\infty \leq \|u\|_\infty \leq C_n.$$

Therefore

$$\int_\Omega g_t \, \mathrm{d}x \geq \frac{\|g_t\|_2^2}{\|g_t\|_\infty} \geq c_n.$$

By the layer-cake formula,

$$\int_\Omega g_t \, \mathrm{d}x = \int_t^M \mu(s) \, \mathrm{d}s.$$

Thus

$$\boxed{\int_t^M \mu(s) \, \mathrm{d}s \geq c_n} \tag{24}$$

for every $t \leq t_0$.

13 Estimate of the truncation level

We can now estimate $\bar{t}$.

Proposition 13.1. *There exist constants $C_n > 0$ and $\delta_0 > 0$ such that, if*

$$\delta \leq \delta_0,$$

then

$$\boxed{\bar{t}^2 \leq C_n \delta.}$$

Proof. Assume first that

$$\bar{t} \le t_0.$$

Since D is nondecreasing, (13) gives

$$\begin{aligned} A - 1 &= 2 \int_0^M D(s)\mu(s)\, \mathrm{d}s \\ &\ge 2 \int_{\bar{t}}^M D(s)\mu(s)\, \mathrm{d}s \\ &\ge 2D(\bar{t}) \int_{\bar{t}}^M \mu(s)\, \mathrm{d}s. \end{aligned}$$

By (22) and (24),

$$A - 1 \ge c_n \bar{t}^2.$$

Using $A - 1 \le \delta$, we conclude that

$$\bar{t}^2 \le C_n \delta.$$

Suppose now that

$$\bar{t} > t_0.$$

Applying exactly the same argument at the fixed level t_0, we obtain

$$A - 1 \ge c_n t_0^2.$$

Since $A - 1 \le \delta$, this is impossible if δ_0 is chosen sufficiently small. Hence, for small deficit,

$$\bar{t} \le t_0,$$

and the previous estimate applies. □

In particular,

$$\bar{t} \le C_n \delta^{1/2}. \tag{25}$$

14 The Faber–Krahn deficit of the core

Recall

$$U = \{u > \bar{t}\}$$

and define

$$g := (u - \bar{t})_+.$$

By construction,

$$|U| = |B_r|.$$

We first compare g with u. Since

$$u - g = \min\{u, \bar{t}\},$$

we have

$$\begin{aligned} \|u - g\|_2^2 &= \int_\Omega \min\{u, \bar{t}\}^2\, \mathrm{d}x \\ &\le \bar{t} \int_\Omega u\, \mathrm{d}x. \end{aligned}$$

Using (17) and (25),

$$\|u - g\|_2^2 \leq C_n \delta^{1/2}. \tag{26}$$

In particular, for small deficit,

$$\|g\|_2$$

is uniformly bounded from below.

We now test

$$-\Delta u = \lambda_1(\Omega) u$$

with g.

On U,

$$u = g + \bar{t}.$$

Thus

$$\begin{aligned} \int_U |\nabla g|^2 \, \mathrm{d}x &= \lambda_1(\Omega) \int_U ug \, \mathrm{d}x \\ &= \lambda_1(\Omega) \left(\int_U g^2 \, \mathrm{d}x + \bar{t} \int_U g \, \mathrm{d}x \right). \end{aligned}$$

Using g as a competitor for $\lambda_1(U)$, we obtain

$$\lambda_1(U) \leq \lambda_1(\Omega) + \lambda_1(\Omega) \bar{t} \frac{\displaystyle\int_U g \, \mathrm{d}x}{\displaystyle\int_U g^2 \, \mathrm{d}x}.$$

The quotient on the right-hand side is uniformly bounded. Therefore

$$\lambda_1(U) \leq \lambda_1(\Omega) + C_n \bar{t}. \tag{27}$$

On the other hand, since

$$|U| = |B_r|,$$

the Faber–Krahn inequality gives

$$\lambda_1(U) \geq \lambda_1(B_r).$$

Therefore

$$\begin{aligned} 0 &\leq \lambda_1(U) - \lambda_1(B_r) \\ &\leq \lambda_1(\Omega) - \lambda_1(B_r) + C_n \bar{t}. \end{aligned}$$

Since $r \leq 1$,

$$\lambda_1(B_r) \geq \Lambda_n,$$

and hence

$$\lambda_1(\Omega) - \lambda_1(B_r) \leq \lambda_1(\Omega) - \Lambda_n = \Lambda_n \delta.$$

Using $\bar{t} \leq C_n \delta^{1/2}$, we obtain

$$\boxed{0 \leq \lambda_1(U) - \lambda_1(B_r) \leq C_n \delta^{1/2}.} \tag{28}$$

Thus a small Kohler–Jobin deficit produces a superlevel set having small Faber–Krahn deficit.

This is the central reduction.

15 Quantitative Faber–Krahn stability

Let

$$\mathcal{R}_D := (-\Delta_D)^{-1}$$

denote the Dirichlet resolvent, with solutions extended by zero outside D.

A quantitative Faber–Krahn estimate for the resolvent gives, for two sets of equal volume,

$$\lambda_1(D) - \lambda_1(B) \geq c_n \inf_{x \in \mathbb{R}^n} \|\mathcal{R}_D - \mathcal{R}_{B(x)}\|^2_{L^\infty \to L^2},$$

up to the appropriate scaling factor.

Applying this to U and B_r, and using (28), we find

$$\inf_{x \in \mathbb{R}^n} \|\mathcal{R}_U - \mathcal{R}_{B_r(x)}\|^2_{L^\infty \to L^2} \leq C_n \delta^{1/2}. \tag{29}$$

Hence

$$\inf_x \|\mathcal{R}_U - \mathcal{R}_{B_r(x)}\|_{L^\infty \to L^2} \leq C_n \delta^{1/4}.$$

This is the first place where the final power 4 appears.

16 Comparison between Ω and the core

We now show that U carries almost all the torsional rigidity of Ω.

Applying the classical Kohler–Jobin inequality to U,

$$\lambda_1(U) T(U)^{\frac{2}{n+2}} \geq \Lambda_n \tau_n^{\frac{2}{n+2}}.$$

Since

$$\lambda_1(U) \leq \Lambda_n + C_n \delta^{1/2},$$

we obtain

$$T(U) \geq \tau_n - C_n \delta^{1/2}.$$

As

$$U \subset \Omega,$$

torsional rigidity is monotone, so

$$T(U) \leq T(\Omega) = \tau_n.$$

Consequently

$$0 \leq T(\Omega) - T(U) \leq C_n \delta^{1/2}. \tag{30}$$

We use the following elementary fact.

Lemma 16.1. *Let $D \subset E$ be bounded open sets. Then*

$$\|\mathcal{R}_E - \mathcal{R}_D\|^2_{L^\infty \to L^2} \leq \frac{T(E) - T(D)}{\lambda_1(E)}.$$

Proof. Let w_E and w_D denote the torsion functions. By positivity and domain monotonicity of the Dirichlet resolvent, for every f with

$$\|f\|_\infty \leq 1,$$

we have

$$|(\mathcal{R}_E - \mathcal{R}_D) f| \leq w_E - w_D.$$

Hence

$$\|\mathcal{R}_E - \mathcal{R}_D\|_{L^\infty \to L^2} \le \|w_E - w_D\|_2.$$

Moreover, extending w_D by zero outside D,

$$\int_E \nabla w_E \cdot \nabla w_D \, \mathrm{d}x = \int_E w_D \, \mathrm{d}x = T(D).$$

Therefore

$$\begin{aligned} \|\nabla(w_E - w_D)\|_2^2 &= T(E) + T(D) - 2T(D) \\ &= T(E) - T(D). \end{aligned}$$

Finally, by Poincaré's inequality on E,

$$\|w_E - w_D\|_2^2 \le \frac{1}{\lambda_1(E)} \|\nabla(w_E - w_D)\|_2^2.$$

Combining the estimates proves the claim. □

Applying Lemma 16.1 to

$$U \subset \Omega$$

and using (30), we obtain

$$\|\mathcal{R}_\Omega - \mathcal{R}_U\|_{L^\infty \to L^2}^2 \le C_n \delta^{1/2}. \tag{31}$$

Thus

$$\|\mathcal{R}_\Omega - \mathcal{R}_U\|_{L^\infty \to L^2} \le C_n \delta^{1/4}.$$

17 Comparison between B_r and B_1

From (5),

$$1 - r \le C_n \delta.$$

Hence

$$T(B_1) - T(B_r) = \tau_n (1 - r^{n+2}) \le C_n \delta.$$

Since

$$B_r(x) \subset B_1(x),$$

Lemma 16.1 yields

$$\|\mathcal{R}_{B_1(x)} - \mathcal{R}_{B_r(x)}\|_{L^\infty \to L^2}^2 \le C_n \delta. \tag{32}$$

Therefore

$$\|\mathcal{R}_{B_1(x)} - \mathcal{R}_{B_r(x)}\|_{L^\infty \to L^2} \le C_n \delta^{1/2}.$$

18 Conclusion

Choose a translation x for which (29) holds. By the triangle inequality,

$$\begin{aligned} \|\mathcal{R}_\Omega - \mathcal{R}_{B_1(x)}\|_{L^\infty \to L^2} \le{}& \|\mathcal{R}_\Omega - \mathcal{R}_U\|_{L^\infty \to L^2} \\ &+ \|\mathcal{R}_U - \mathcal{R}_{B_r(x)}\|_{L^\infty \to L^2} \\ &+ \|\mathcal{R}_{B_r(x)} - \mathcal{R}_{B_1(x)}\|_{L^\infty \to L^2}. \end{aligned}$$

Using (29), (31), and (32),

$$\inf_{x \in \mathbb{R}^n} \|\mathcal{R}_\Omega - \mathcal{R}_{B_1(x)}\|_{L^\infty \to L^2} \le C_n \delta^{1/4}.$$

Equivalently,

$$\boxed{\delta \geq c_n \left(\inf_{x \in \mathbb{R}^n} \|\mathcal{R}_\Omega - \mathcal{R}_{B_1(x)}\|_{L^\infty \to L^2} \right)^4 .} \tag{33}$$

Returning to the original scale, let B_T be a ball satisfying

$$T(B_T) = T(\Omega)$$

and denote its radius by R_T. Since

$$\|\mathcal{R}_{rD}\|_{L^\infty \to L^2} = r^{\frac{n}{2}+2} \|\mathcal{R}_D\|_{L^\infty \to L^2},$$

the scale-invariant form of (33) is

$$\boxed{\delta_{\mathrm{KJ}}(\Omega) \geq c_n \left[R_T^{-\frac{n+4}{2}} \inf_{x \in \mathbb{R}^n} \|\mathcal{R}_\Omega - \mathcal{R}_{B_T(x)}\|_{L^\infty \to L^2} \right]^4 .}$$

19 Where the exponent is lost

The origin of the fourth power can now be seen explicitly.

First,

$$A - 1 \lesssim \delta.$$

The level-set identity and the quadratic growth of D yield

$$\bar{t}^2 \lesssim \delta,$$

hence

$$\bar{t} \lesssim \delta^{1/2}.$$

The Faber–Krahn deficit of the core then satisfies

$$\lambda_1(U) - \lambda_1(B_r) \lesssim \delta + \bar{t} \lesssim \delta^{1/2}.$$

Finally, quantitative Faber–Krahn stability gives

$$d_{\mathrm{res}}(U, B_r)^2 \lesssim \lambda_1(U) - \lambda_1(B_r),$$

and consequently

$$d_{\mathrm{res}}(U, B_r) \lesssim \delta^{1/4}.$$

Thus

$$\delta \gtrsim d_{\mathrm{res}}^4.$$

The loss therefore does not come from the quantitative Faber–Krahn theorem itself. It occurs earlier, in the estimate

$$\bar{t}^2 \lesssim \delta.$$

A sharp quadratic resolvent stability estimate would require a mechanism avoiding this square-root loss.

20 Relation with the quantitative Chiti argument

The structure is now transparent.

In a quantitative Chiti argument one has schematically

$$\text{single crossing} + \text{norm deficit} \longrightarrow \text{control of the rearrangements},$$

which is then converted into

$$\bar{t} \text{ small} \longrightarrow U_{\bar{t}} \text{ almost optimal for Faber–Krahn}.$$

In the present Kohler–Jobin argument, the first part is replaced by

$$\text{modified torsion} + \text{Kohler–Jobin rearrangement} \longrightarrow A - 1 \leq \delta,$$

followed by the exact identity

$$A - 1 = 2 \int_0^M D(t)\mu(t)\, \mathrm{d}t.$$

Thus the correspondence is

$$\boxed{\text{control of } u^* - z^* \quad \longleftrightarrow \quad A - 1 = 2 \int D\mu.}$$

After the control of the levels has been obtained, both arguments follow the same basic mechanism:

$$\boxed{\text{level control} \longrightarrow \bar{t} \text{ small} \longrightarrow U_{\bar{t}} \text{ has small FK deficit} \longrightarrow \text{quantitative Faber–Krahn stability.}}$$

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